

MEMBER 1: BACKEND & ARCHITECTURE

Focus: Model and Controller Layers (MVC), API Design, and Real-time Logic.

| Primary Responsibilities

- **Complete CRUD Operations:** Finalize controllers for `Users`, `Posts`, and `Groups`. Ensure every model supports Create, Read, Update, and Delete.
- **Search Functionality (Req #10):** Implement server-side filtering logic for posts (by content/team) and users (by username).
- **Access Control Logic:** Ensure a user can only delete/edit their own posts and that `isAdmin` (Group Managers) have unique permissions to approve requests.
- **Socket.io Server Management:** Handle the backend "Rooms" logic to ensure team-specific chat messages are broadcast only to the correct participants.
- **Database Seeding (Req #49):** Create the `/api/seed` endpoint to populate the network with dummy data for the project defense.

| JSON Format Requirements (Input/Output)

To ensure the Frontend doesn't crash, Member 1 must ensure all endpoints return this exact structure:

```
// User Object (Output for Login/Me)
{
  "_id": "string",
  "username": "string",
  "favoriteTeam": "string",
  "isAdmin": boolean,
  "groups": ["id_1", "id_2"],
  "createdAt": "ISO_Date_String"
}

// Post Object (Output for Feed)
{
  "_id": "string",
  "content": "string",
  "author": { "_id": "id", "username": "name" }, // Populate this!
  "likes": ["user_id_1", "user_id_2"],
  "teamTag": "string",
}
```

```
"createdAt": "ISO_Date_String"  
}
```

Critical: Always use `_id` (with underscore) to match MongoDB Atlas!